

Radiator Sustainability Final Report
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Description of Product

During the operation of a passenger vehicle, waste heat is generated in the drive unit, engine, electrical systems, and passenger compartment. This heat must be rejected to the ambient air, typically through the use of a water to air heat exchanger called a radiator. The radiator is ubiquitous in water-cooled cars, trucks, motorcycles, and heavy equipment whether or not they are ICE or electric powered. They play a key role in the cooling system of these vehicles and will continue to be necessary as transportation transitions to electric vehicles.

The radiator prevents overheating through effective heat dissipation, maintaining the engine temperature within safe limits, preventing overheating and potential engine damage. It also improves performance because a well-regulated engine temperature ensures optimal combustion and engine efficiency, contributing to better fuel economy and performance. Lastly the radiator extends engine life by preventing thermal stress and wear on engine components, enhancing the longevity of the engine.

The standard construction of a radiator includes a core made of tubes and fins, and inlet/outlet end tanks. Coolant flows through the tubes, and the fins between each tube increase the surface area for heat transfer to the ambient air. The tubes and fins are generally constructed from aluminum or copper. On either side of the tubes, there are end tanks - an inlet and outlet tank - that aim to uniformly distribute the flow through the core to maximize the efficiency of heat transfer. These end tanks are generally either plastic or aluminum. There are additional parts such as rubber gaskets, aluminum header plates, side plates, and radiator caps that are also included in the final application. This analysis will be focused on an aluminum core (tubes, fins, and headers), and Nylon 6 plastic end tanks.

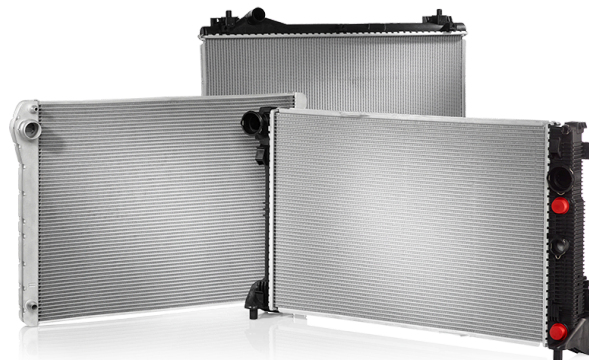


Figure 1: Automotive Radiators (Nissens)

Source: <https://nissens.com/en-us/engine-cooling/radiators.aspx>

Motivation

The global car radiator market size was \$4.98 Billion as of 2023, and is projected to grow at a compound annual growth rate of 5.9% to a market size of \$8.9 Billion in 2033. (FactMR).

Vehicle production globally is increasing, and will continue to increase over the next decade. The growth of the market for radiators will follow this trend, and the increase in demand for hybrid and electric cars will further spur demand because electric batteries and motors require substantial cooling (FactMR). If the environmental efficiency of the life cycle of a radiator could be improved, there would be large benefits due to the market size of the product.

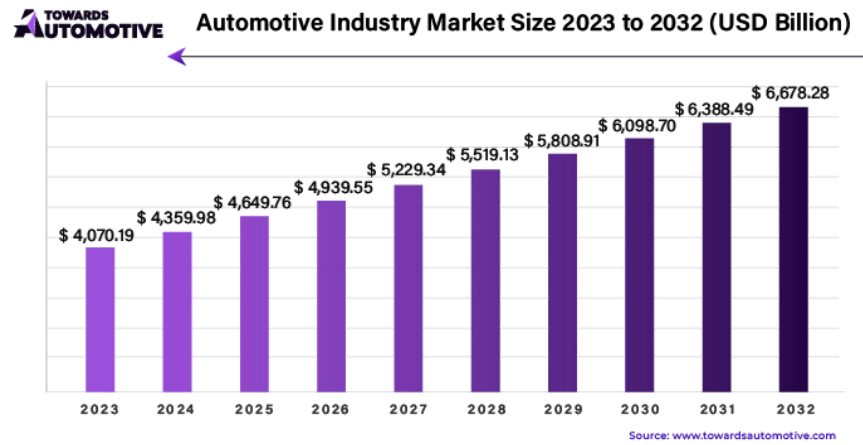


Figure 2: Global Automotive Market Trends (Towards Automotive)

Source: www.towardsautomotive.com

Scope of Production and Analysis

The radiator life cycle is comprised of the following steps:

- Raw material production (aluminum, plastic)
- Radiator manufacturing (tube, fin, tanks, and assembly procedure)
- Use phase (installation in a passenger vehicle)
- End-of-life (recycling or scrap)

Current Production

The core of the radiator - composed of tubes, fins, and headers - are first individually manufactured. As seen in Figure 6, these components start as the same raw material, coiled aluminum. Different deformation processes are used to achieve the final desired shape of each. For the tubes, the coiled aluminum is folded by a machine to form it into a tube, then it is welded to seal it and cut to length (Foshan Sunhope Co). A sample of the tubes will be leak tested to ensure they won't leak during operation. The cross section can be seen in Figure 3 below. More complicated geometries may have multiple tube sections or dimples that increase heat transfer by introducing turbulence in the flow. This analysis considers a simple flat sided aluminum radiator tube as shown.

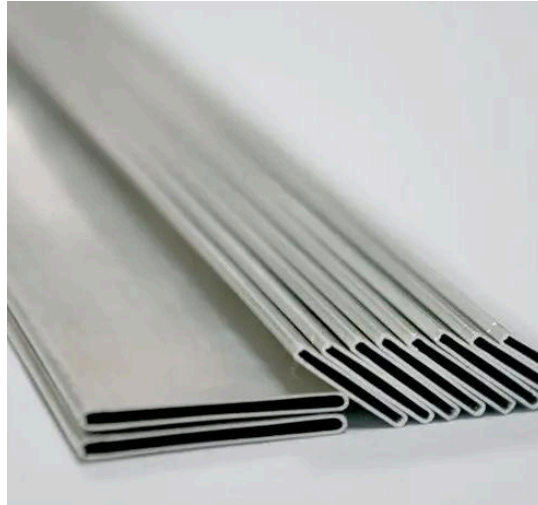


Figure 3: Aluminum Radiator Tubes (Chaluminum)

Source:

<https://www.chaluminium.com/the-oval-advantage-how-flat-sided-aluminum-tubes-boost-radiator-performance>

The header plates are also manufactured from coiled aluminum. They are first cut to length by a shearing machine, then are stamped and punched in a die to the desired final shape shown below in Figure 4. In the final assembly, the tubes will be inserted into the headers to hold them together and maintain their spacing (Foshan Sunhope Co).



Figure 4: Radiator Header Plates

Source:

https://www.made-in-china.com/video-channel/njmjst_ZjRQLerYYFcp_Different-Types-of-Aluminum-Header-Plates-for-Radiators.html

The radiator fins start as a coiled aluminum, like the other parts of the radiator core. The fins are formed, and compressed to hold their shape, then are cut to length. The length of the fins and

tubes will depend on the final dimensions of the radiator, and will vary depending on the product application. The radiator width, and numbers of tubes and therefore fins will vary between different types of vehicles so each radiator will have a different geometry.

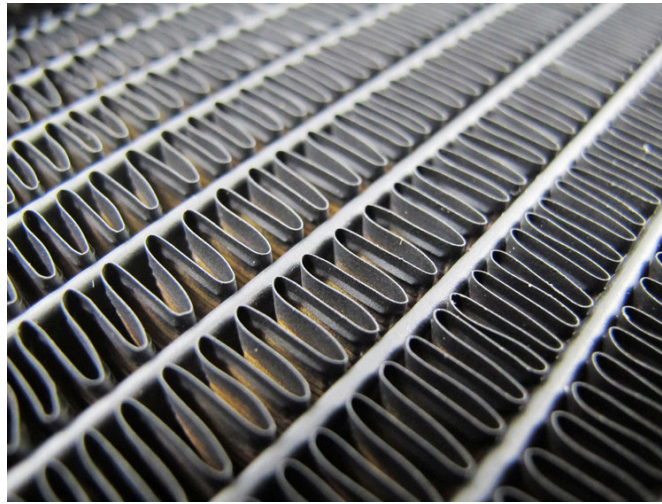


Figure 5: Radiator Fins

Source: <https://www.xs-pc.com/radiators-rs-series/rs120-single-fan-radiator>

Manufacturing Process Flow Chart

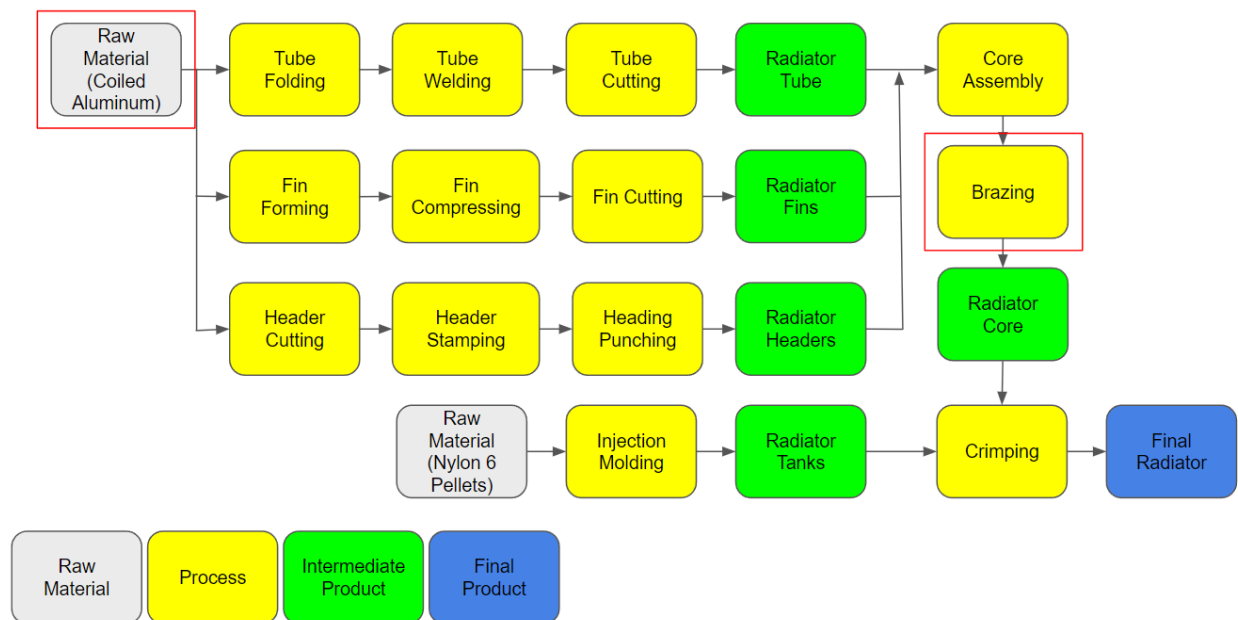


Figure 6: Radiator Production Process

Assembly Process

After the tubes, fins, and headers are individually manufactured, the core will be assembled and brazed, as shown in Figure 6 above. This process may be machine automated, but this analysis will assume that a human operator is assembling the core with the help of fixturing. The tubes

and fins are alternated, and the headers are inserted onto the ends of the tubes. This core assembly is then placed into a brazing oven that will apply flux and heat, after which it is a complete core.

Radiator Tanks Manufacturing

The radiator tanks are manufactured by an injection molding process from PA-6 material as shown in figure 6 above.

Final Assembly Process

In the final assembly, the brazed radiator cores and the inlet and outlet tanks are joined together. The inlet and outlet tanks are placed into the headers, and tabs formed onto the headers are crimped to keep the inlet and outlet tanks in place. This is the final assembly step of the radiator.

Table 1: Streamlined LCA of the current production of automotive radiators. This analysis assumes that the core material is aluminum, it is transported from the country of origin by ship and truck to the final destination, and is recycled at the end of the functional life.

Automotive Radiator					
	Material	Manufacture	Transport	Use	Disposal
Material resources (high use = 0, none = 4)	1	2	2	4	3
Energy use (high use = 0, none = 4)	1	1	1	4	3
Global warming (much CO2 = 0, no CO2 = 4)	2	1	1	4	2
Human health (Toxic emissions or waste ?)	2	2	1	4	3
The biosphere (large impact = 0, no impact = 4)	2	2	1	4	3
Column totals	8	8	6	20	14
				Sum across column totals	56

For the streamlined LCA, the aluminum material, manufacture, and transport are high intensity categories. For this analysis, the radiators are manufactured in Mexico from aluminum sourced from China, and delivered by ship and truck to the OEM in the USA. The aluminum material is virgin non-recycled material, and the manufacturing site uses substantial amounts of electricity from fossil fuel sources. The methods of shipping across large distances may be economically sustainable but are more environmentally costly than sourcing materials closer to the production facility or final end-use destination in the USA. During the use phase, the radiator has low

impact because it is a passive heat transfer device. When it is disposed of, the aluminum core is readily recycled and the plastic tanks may be recycled as well.

Economic Sustainability

It is important for both the supplier and car manufacturer to maintain a profit margin that allows them to stay in business. In 2022, the average profit margin for an automotive OEM was 8.3% (Bain). All components of the car, including the radiator must be manufactured cost-effectively to maintain a profit margin. Any proposed change to improve sustainability must not substantially increase the cost, or may even lower the cost per unit.

Alternatives for Higher Sustainability

First Alternative

Manufacturers can consider different materials to improve the efficiency of the embodied energy, CO2 equivalent, and water usage for the radiator. The embodied energy is all of the energy associated with manufacturing a product. It can be used to make comparisons between two different material selections, along with other important information such as material density, cost, and in this case, thermal conductivity, which is an important figure for a heat exchanger. For radiators, aluminum is dominating the market but copper is also a popular choice that can be considered. The table below shows the comparison between aluminum and copper alloys in some key metrics (Ashby, 2021). Copper heat exchangers will be soldered with lead, so there will be a larger externality of impact on the workers involved in making the radiators.

Table 2: Aluminum vs Copper Material Properties (Ashby, 2021)

Material	Aluminum	Copper
Density	2800 kg/m ³	8300 kg/m ³
Price*	\$4.3 / kg	\$5.6 / kg
Thermal Conductivity**	237 W/m/K	398 W/m/K
Embodied Energy	190 MJ/kg	59 MJ/kg
CO2 Equivalent	12 kg/kg	3.6 kg/kg
Water Usage	1000 L/kg	310 L/kg
Recycle Fraction	43%	43%
R-Energy	33 MJ/kg	13 MJ/kg
R-carbon CO2 Eq	2.6 kg/kg	1.1 kg/kg

According to data from Ashby, copper has a lower embodied energy, CO2 equivalent, and water usage per kg. It also has a higher thermal conductivity than aluminum, making it more effective at transferring heat, so less material may be used to achieve the same heat transfer rate. Copper is also highly recyclable, equivalent to aluminum (Aluminum v Copper, 2018). According to Ashby, the recycle fraction is comparable between the two materials but copper requires less energy and CO2 equivalent to recycle, meaning it may be environmentally preferable in the end-of-life stage. The higher price and increased material density mean the part will be heavier and more expensive. A heavier part leads to a heavier car, which will be slightly less efficient. Manufacturers will need to perform feasibility studies (not part of this report) to determine if the higher cost of copper can be justified.

Second Alternative

Figure 7 below shows the unit manufacturing process for brazing, a hotspot process for energy use in the radiator manufacturing industry. This process requires chemical or electrical energy, potentially combustion of fossil fuels, and can generate pollutants (EAA Aluminum Automotive Manual).

A more efficient brazing oven could be used to reduce the required inputs of energy (natural gas) and reduce the waste outputs (pollutants, heat). According to a paper from Arnal et al, 2020, heat regeneration systems can be retrofit to existing furnaces to increase the efficiency of natural gas combustion and reduce CO2 and NOx emissions by using oven exhaust gasses to preheat incoming combustion air, which can be seen in Figure 8 below.

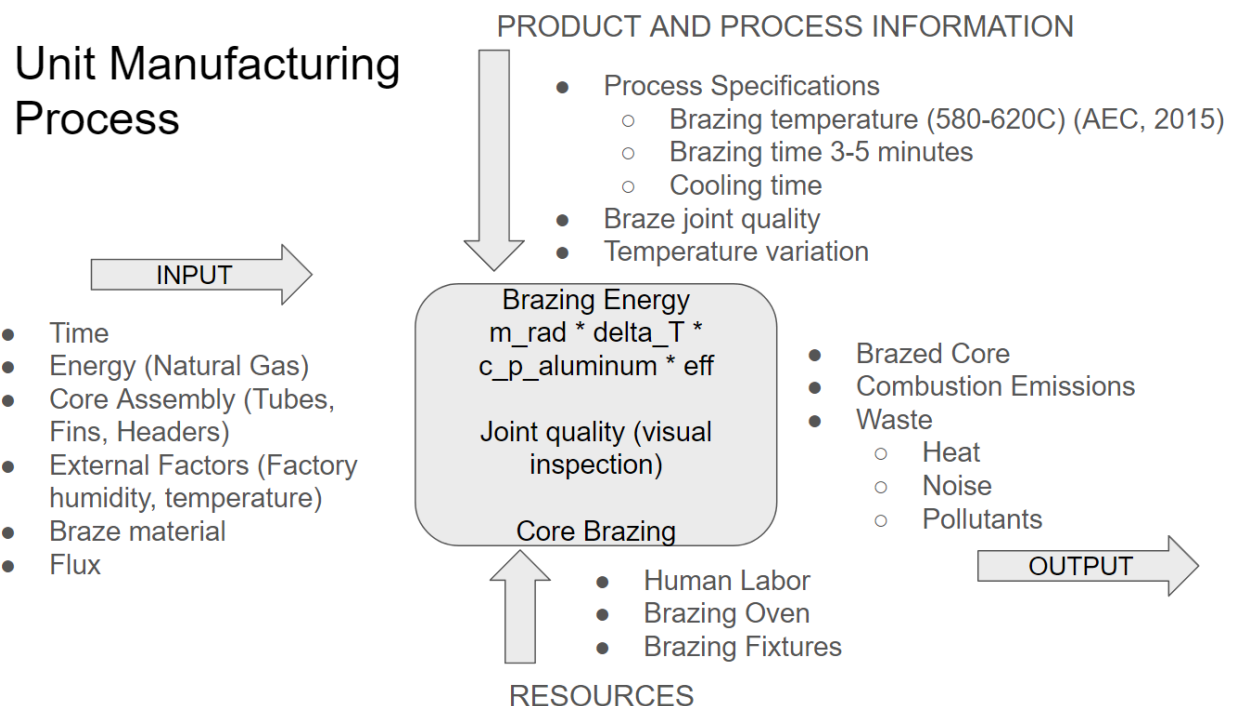


Figure 7: Unit Manufacturing Process for Brazing

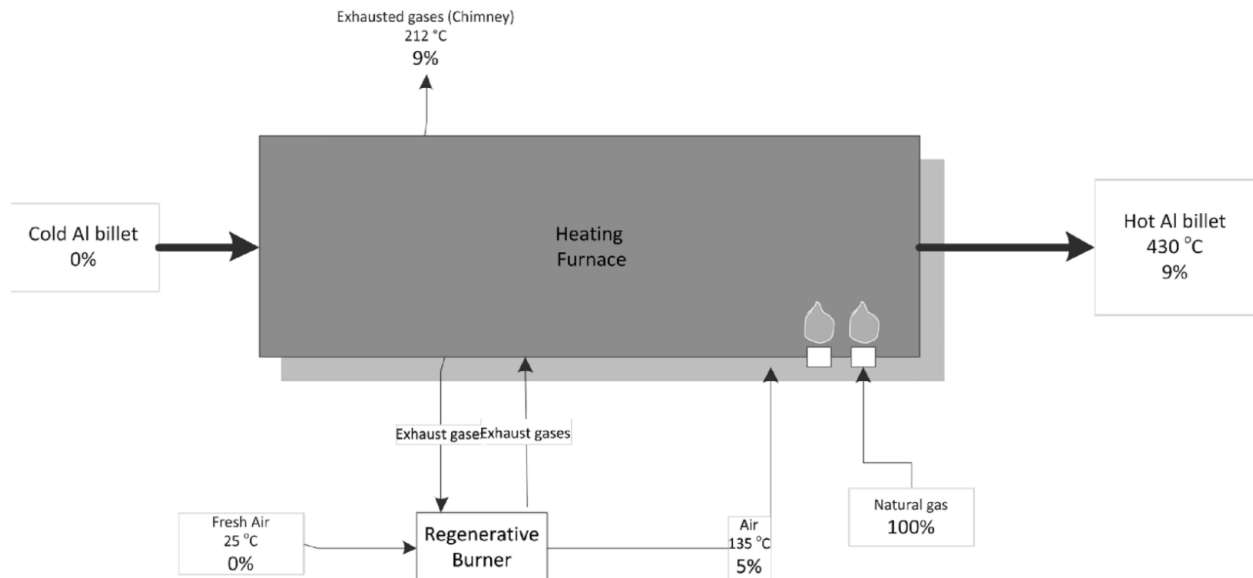


Figure 8: Regenerative Burner Schematic

Table 3 below shows the improvements found by the energy recovery systems that can be retrofitted to existing heating furnaces or brazing ovens. The economic payback period and feasibility is not included in this report (Arnal et al., 2020). There are potential natural gas, CO₂, and NO_x emissions reductions as a result of this retrofit.

Table 3: Retrofitting Analysis for Heating Furnace

Domain		Technical		Environmental		TA
Innovation	MY (%)	Volume NG saved/mass Al (Nm ³ /kg)	CO ₂ emissions (gCO ₂ eq/kg Al saved)	NO _x emissions gNO _x eq/kg Al saved		Exergy cost reduction (Wh/kg Al)
Energy recovery	10	0.001	26	2.706		78
Refractories	37	0.098	198	20.040		365
Co-firing	–	0.039	78	8.119		–

Summary and Conclusions

Radiators are a ubiquitous product that are currently being made from aluminum and plastic commonly in Mexico and China for OEMs all over the world. There are some improvements that can be studied in the radiator industry that could reduce the embodied energy, CO₂, and NO_x emissions of the various processes of the manufacturing process. Because they are material heavy, during their manufacture and transport, they have large economic impacts. By potentially using more recyclable materials, or those with lower embodied energy (copper v aluminum), the radiators can be made more environmentally sustainable. In the facilities where the radiator cores are brazed, retrofit improvements could be made to older brazing ovens that would recapture heat from exhaust gasses, reducing the volume of natural gas that needs to be burned in these facilities.

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